



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Year: 2026 Trimester: Spring

CSE 4165/CSE 465 Web Programming

Total Marks: 30

Duration: 2 Hours

SET - B

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Create a JavaScript program where a user enters a patient's vitals and your code shows a live health report. Strategies are as follows:

[10]

Index.html:

Create two text fields (input type="text") to take the user's Heart Rate (bpm) and Oxygen Saturation / SpO2 (%), and a "Save Reading" button. Below the button there is a label field to show the report and feedback.

Script.js:

1. Take the user's Heart Rate and SpO2 from the input fields in the HTML file.
2. Keep track of every reading entered so far (for example, in arrays).
3. Inside a function:
 - a) Calculate and display the Average Heart Rate (of all readings entered so far, including the current one).
 - b) Calculate and display the Average Oxygen Saturation (of all readings so far, including the current one).
 - c) Calculate and display the Risk Score from the current reading using the formula:
$$\text{RiskScore} = (100 - \text{SpO2}) \times 2 + |\text{HeartRate} - 80| \times 0.5$$

($|\text{HeartRate} - 80|$ is the absolute / positive difference, e.g. $|78 - 80| = 2$.)
 - d) Display the feedback in the label area: "SAFE" if the Risk Score is 10 or less, "WARNING" if it is greater than 10 and 20 or less, and "DANGER" if it is greater than 20.

Sample Input / Output

Sample Input	Sample Output	
Heart Rate, SpO2 (entered in order)	Report	Feedback
80, 97	Average Heart Rate: 80 Average Oxygen Saturation: 97 Risk Score: 6.0	SAFE
78, 95	Average Heart Rate: 79 Average Oxygen Saturation: 96 Risk Score: 11	WARNING
130, 80	Average Heart Rate: 96 Average Oxygen Saturation: 90.67 Risk Score: 65	DANGER

2. The UIU Mars Rover Team is preparing for a rover field-test mission and needs to calculate how many battery modules must be carried for the rover. Write a PHP program that takes three inputs from an HTML form: the mission distance in meters, the energy consumption per meter, and the capacity of one battery module.

[10]

The program should first calculate the total energy required by multiplying the mission distance with the energy consumption per meter. Since the team cannot carry partial battery modules, the program must calculate the minimum number of complete battery modules needed. Then, calculate the unused energy capacity after the required number of battery modules is carried.

Classify the battery usage efficiency using the following conditions:

- Efficient: unused energy is less than or equal to 10% of total carried energy.
- Acceptable: unused energy is more than 10% but less than or equal to 25% of total carried energy.

- **Wasteful:** unused energy is more than 25% of total carried energy

Finally, display the **total energy required**, **number of battery modules needed**, **unused energy capacity**, and the **battery usage status**.

Sample Input taken using HTML form			Sample Output printed using PHP code			
Mission Distance	Energy per Meter	Battery Capacity	Total Energy Required	Battery Modules	Unused Energy	Status
1200	8	5000	9600	2	400	Efficient
1500	9	4000	13500	4	2500	Acceptable
1000	6	5000	6000	2	4000	Wasteful

Create a database named 'uiu_bookshop' and a table named 'book_info' with the following structure and values:

[10]

BookID	BookTitle	Author	Category	Price	Stock
1	Web Basics	John Smith	Programming	450	10
2	Database Guide	Alice Roy	Programming	550	8
3	English Grammar	Mary Khan	Language	300	15
4	Business Math	David Lee	Business	400	12
5	Python Mastery	Sara Ahmed	Programming	600	5
6	French Made Easy	Paul Costa	Language	350	7

Now, do the following (write full PHP-MySQL code):

1. Display all books from the Programming category, sorted by Price in descending order.
2. Display the BookTitle, Author, and Price of all books where the Price is greater than 400 BDT and the Category is 'Programming'.
3. Display the number of books available in each Category.
4. Display the total net worth of the entire bookshop. (Hint: To find the total net worth of the entire bookshop, multiply the Price and Stock of each book, then add all the values together.) 23680
5. Apply a 10% price reduction to all books in the 'Language' category, then display the BookTitle and updated Price of those books.

$$\frac{450 \times 10}{1000}$$

$$1000 \times 19$$

$$12000 (450 \times 10) + (550 \times 8)$$

$$4500 + 4400$$